THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $\frac{2}{3}$ of a number is 14, the number is: 1

(a) 42 **(b)** 21 **(c)** $9\frac{1}{3}$ **(d)** $10\frac{1}{2}$

2. An angle is 30° less than its supplement. The angle is: 1

(a) 65° **(b)** 75° **(c)** 85° **(d)** 95°

3. The sum of 5 consecutive numbers starting at 10 is: 1

(a) 62 **(b)** 60 **(c)** 61 **(d)** 59

4. The angles of a triangle are $(60 - d)^\circ$, 60° and $(60 + d)^\circ$. When $d = 10$, the largest angle is: 1

(a) 90° **(b)** 60° **(c)** 70° **(d)** 80°

5. A tap fills $\frac{2}{5}$ of a tank in 1 hour. The time (hours) to fill the whole tank is: 1

(a) 5 **(b)** $2\frac{1}{2}$ **(c)** $2\frac{5}{6}$ **(d)** $1\frac{1}{4}$

6. A transversal makes alternate interior angles of 65° and 65°. The two lines are: 1

(a) perpendicular **(b)** parallel **(c)** intersecting **(d)** not parallel

7. For every whole number n, $n^2 + n$ is always: 1

(a) odd **(b)** a prime number **(c)** even **(d)** a multiple of 3

8. In two triangles, $AB = PQ$, $BC = QR$, $\angle B = \angle Q$. They are congruent by: 1

(a) not congruent **(b)** SAS **(c)** ASA **(d)** SSS

9. Which is the greatest? 1

(a) $\frac{3}{4} \div \frac{1}{8}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{8}$ **(d)** $\frac{3}{4} \times \frac{1}{8}$

10. Two supplementary angles differ by 20°. The larger angle is: 1

(a) 90° **(b)** 110° **(c)** 120° **(d)** 100°

11. A number has digit sum 12. It is divisible by: 1

(a) 3 but not 9 **(b)** both 3 and 9 **(c)** neither 3 nor 9 **(d)** 9 but not 3

12. An isosceles triangle has equal sides 9 cm and base 6 cm. Its perimeter (cm) is: 1

(a) 26 **(b)** 25 **(c)** 24 **(d)** 23

13. A car travels $\frac{35}{2}$ km on 1 litre of petrol. On 6 litres it travels (km): 1

(a) $52\frac{1}{2}$ **(b)** 105 **(c)** $105\frac{1}{2}$ **(d)** 210

14. Two complementary angles are in the ratio $1:2$. The larger angle is: 1

(a) 80° **(b)** 70° **(c)** 60° **(d)** 50°

15. Adding 72 and its reverse gives: 1

(a) 99 **(b)** 100 **(c)** 98 **(d)** 101

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. The sum of three consecutive odd numbers is 51. Find them. 2

17. The exterior angle of a triangle is $(4x - 10)^\circ$ and the interior opposite angles are $(x + 12)^\circ$ and $(2x + 18)^\circ$. Find x and the exterior angle. 2

18. $\frac{3}{4}$ of a number is 27. Find the number and also $\frac{1}{4}$ of it. 2

19. A transversal cuts two lines and makes alternate interior angles of 80° and 80°. Are the lines parallel? Give a reason. 2

20. Use reverse-and-add on 39 and state how many steps are needed to get a palindrome. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Mohan has ₹240. He spends $\frac{1}{4}$ of it and shares the rest equally among 6 children. How much does each child get? 3

22. Two supplementary angles are in the ratio $1:3$. Find both angles and the complement of the smaller one. 3

23. Write 75 as the sum of 5 consecutive numbers. 3

24. $\triangle ABC \cong \triangle DEF$ with $AB = (5x + 3)$ cm, $DE = (2x + 18)$ cm, $BC = 7$ cm and $AC = 9$ cm. Find x, the length DE and the perimeter of $\triangle DEF$. 3

OR

Evaluate $\left(\frac{1}{2} + \frac{1}{3}\right) \times \frac{6}{5}$. 3

25. Two lines intersect and two adjacent angles are in the ratio $1:8$. Find all four angles. 3

OR

Apply Kaprekar's routine to 9810 repeatedly until you reach 6174. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The four angles around a point are in the ratio $1:2:3:4$. Find the angles and say which of them are acute, right or obtuse. 5

OR

List all three-digit palindromes with first digit 1 that are divisible by 3. 5

27. Four sticks of lengths 2 cm, 4 cm, 6 cm and 7 cm are available. Choosing any three, how many different triangles can be made? Test each choice. 5

OR

A jug holds $2\frac{1}{2}$ litres of squash. $\frac{2}{5}$ of it is syrup and the rest is water. Find the syrup and the water. How many glasses of $\frac{1}{4}$ litre can be filled from the jug? 5

END OF QUESTION PAPER — **Total: 50 Marks**